NODES ARE STABLE UNDER ETALE MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-curves/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction step invokes \label{lemma-node-field-extension} (source line 4669), a published lemma of the same chapter whose own proof delegates its central ring-isomorphism step to the reader (source lines 4646-4652); that lemma was therefore not carded separately, but it is used here as published input per the citation policy; the completion isomorphism and the final conclusion are complete in the present proof.

Editorial note. The label lemma-node-etale-local names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Nodes under field extension. Same chapter, Lemma \label{lemma-node-field-extension}. Proof lines 4669-4671: one may base change to the algebraic closure of k, where the residue fields of x and y are both k.
- Completions of etale local rings. Etale Morphisms, Lemma \label{etale-lemma-characterize-etale-completions}; More on Algebra, Lemma \label{more-algebra-lemma-flat-unramified}. Proof lines 4673-4678: the map on completed local rings O_{X,x}^ -> O_{Y,y}^ is an isomorphism.
- The definition of a node via the completed local ring. Same chapter, Lemma \label{lemma-nodal}, Lemma \label{lemma-nodal-algebraic}. Proof lines 4678-4679: being a node is expressed through the completed local ring (the normal form kappa[[s, t]]/(st)), so an isomorphism of completions preserves it.

Source reference. Original lemma 039M: sources/stacks-chow/tags.txt, source lines 4277–4277. Prerequisite pointer invoked at proof line(s) 4678; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C56: sources/stacks-curves/pinned-original-source.tex, source lines 4605–4605. Prerequisite pointer invoked at proof line(s) 4673; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AGX: sources/stacks-chow/tags.txt, source lines 13210–13210. Prerequisite pointer invoked at proof line(s) 4679; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let k be a field. Let X be a locally algebraic k-scheme of dimension 1. Let $Y \to X$ be an étale morphism. Let $y \in Y$ be a point with image $x \in X$. The following are equivalent*

(1) *x is a closed point of X and a node, and*
(2) *y is a closed point of Y and a node.*

Proof. By Lemma 0C56 we may base change to the algebraic closure of k. Then the residue fields of x and y are k. Hence the map $\mathcal{O}^\wedge_{X,x} \to \mathcal{O}^\wedge_{Y,y}$ is an isomorphism (for example by Étale Morphisms, Lemma 039M or More on Algebra, Lemma 0AGX). Thus the lemma is clear. □